

# Conditional Characterization of RG Admissibility Protocols

## Coherence Thresholds, Robustness Radius, Generated Closures, and the Remaining Transport Modulus

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August 6, 2026

### Abstract

Resolution Geometry is fibrewise canonical once a declared admissible experiment  $(\mathcal{E}, \Gamma)$  is fixed, but the protocol  $\Gamma = (G, \tau, N, S, C, T)$  is not selected by observation alone. We characterize how much of this residual protocol structure becomes canonical after its irreducible operational generators are declared. A local decision or resource functional determines the comparison metric as its Hessian. Continuous positive compositional coherence forces an exponential semigroup, and multiplicative survival has a unique additive coordinate  $-a \log c$  up to scale. Naturality forces coherence loss into spectral functional calculus of the information operator  $L = G^{-1}F$ . Positive homogeneity leaves inverse powers; an explicit reciprocity axiom  $\lambda\psi(\lambda) = a$  uniquely selects  $K = aL^{-1}$  on the positive information sector. A fixed coherence horizon and survival budget then determine a unique maximal hard sector. Stability is characterized by distance to the complete declared discriminant rather than by an arbitrary condition-number cutoff. Declared nuisance generators determine a unique smallest saturated equivalence relation, and available garbling generators determine a unique maximal composition-closed information-order ideal. Metric and projector compatibility do not, however, select a unique connection: transport and holonomy retain genuine moduli unless an additional variational or geometric principle is supplied.

A deterministic finite battery supports each boundary: the inverse constitutive law is the sole survivor of the tested natural, homogeneous, antitone, reciprocal family; nuisance and coarse-graining closures are unique; distance to the discriminant has the required boundary properties; and four distinct compatible transport generators survive. The result conditionally closes the structural form of  $G, \tau, N, S$ , and  $C$ , while isolating connection selection as the remaining intrinsic protocol problem. It does not derive losses, resources, gauge generators, budgets, physical channels, or dynamics from observation alone.

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## 1 Introduction

The earlier characterization of RG established the logical chain

$$(\mathcal{E}, \Gamma) \longrightarrow Q_{\mathcal{E}} \longrightarrow g_F \longrightarrow P_{\tau} \longrightarrow W_{\text{obs|adm}} \longrightarrow \{T_{ij}\}, \quad (1)$$

and fixed the resulting geometry up to protocol-preserving natural isomorphism. The remaining foundational question is not whether the bare experiment determines  $\Gamma$ ; it does not. The question is whether each component of  $\Gamma$  has a canonical construction after its minimal operational generators are declared.

We prove a conditional characterization. In each case the distinction is explicit:

- external data declare a loss, resource, gauge generator, uncertainty budget, channel generator, or connection-selection principle;
- closure, differentiation, spectral calculus, maximality, and naturality determine the resulting RG protocol component.

The result is analogous to other mature geometries: an input structure is not selected from nothing, but its natural consequences are canonical.

## 2 Protocol data and operational generators

**Definition 2.1** (Admissibility protocol). *An RG admissibility protocol is*

$$\Gamma = (G, \tau, N, S, C, T), \quad (2)$$

where  $G$  is a comparison metric,  $\tau$  a hard or soft resolution rule,  $N$  nuisance closure,  $S$  stability and conditioning,  $C$  admissible coarse-graining, and  $T$  transport and gluing data.

**Definition 2.2** (Operational generator package). *A generator package is*

$$\Xi = (\mathcal{R}, \mathcal{B}, \mathcal{H}_N, \mathcal{U}, \mathcal{M}, \mathcal{V}), \quad (3)$$

where  $\mathcal{R}$  is a local decision/resource functional,  $\mathcal{B}$  a coherence normalization and budget,  $\mathcal{H}_N$  declared nuisance generators,  $\mathcal{U}$  an uncertainty/perturbation class,  $\mathcal{M}$  available garbling generators, and  $\mathcal{V}$  optional connection-selection data.

**Remark 2.3.** *The theorem characterizes  $\Gamma$  from  $(\mathcal{E}, \Xi)$ , not from  $\mathcal{E}$  alone. Removing  $\Xi$  restores the protocol freedom demonstrated by earlier countermodels.*

## 3 Comparison metric

**Assumption 3.1** (Local resource regularity). *Let  $\mathcal{R}$  be a twice differentiable local loss or resource functional on the regular observable tangent domain, strictly convex at the operating point and additive on independent products.*

**Definition 3.2** (Resource comparison metric). *Define*

$$G = \nabla^2 \mathcal{R}. \quad (4)$$

**Theorem 3.3** (Conditional characterization of  $G$ ). *For a fixed twice differentiable local resource functional  $\mathcal{R}$ , the unique symmetric bilinear form representing its second variation is  $G = \nabla^2 \mathcal{R}$ . Strict convexity makes  $G$  positive definite. Covariance follows from the tensorial transformation of the Hessian at a critical operating point, and product additivity gives direct-sum composition.*

*Proof.* The second-order Taylor coefficient of  $\mathcal{R}$  is unique. Strict convexity gives positivity. Under an admissible coordinate transformation at the operating point, the second differential is a covariant bilinear form. If  $\mathcal{R} = \mathcal{R}_1 + \mathcal{R}_2$  on a product, mixed second variations vanish and the Hessian is  $G_1 \oplus G_2$ .  $\square$

**Non-claim 3.4.** *Observation does not select  $\mathcal{R}$ . The result characterizes  $G$  after a decision or resource geometry is declared.*

## 4 Coherence clock and threshold

**Axiom 4.1** (Coherence semigroup). *On a regular observable sector, coherence survival is represented by a strongly continuous positive contraction semigroup  $(C_t)_{t \geq 0}$  satisfying*

$$C_0 = I, \quad C_{t+s} = C_t C_s. \quad (5)$$

**Theorem 4.2** (Coherence generator). *In finite dimension, there exists a generator  $K$  such that*

$$C_t = e^{-tK}. \quad (6)$$

*If the semigroup is self-adjoint and contractive, then  $K \succeq 0$ .*

*Proof.* This is the standard finite-dimensional one-parameter semigroup representation. Self-adjoint contractivity places the spectrum of  $C_t$  in  $(0, 1]$ , hence the logarithmic generator is positive semidefinite.  $\square$

**Theorem 4.3** (Additive coherence time). *Let  $h : (0, 1] \rightarrow [0, \infty)$  be continuous, strictly decreasing, satisfy  $h(1) = 0$ , and convert multiplicative survival into additive duration:*

$$h(xy) = h(x) + h(y). \quad (7)$$

*Then*

$$h(x) = -a \log x \quad (8)$$

*for a unique  $a > 0$ .*

*Proof.* Set  $x = e^{-u}$  and define  $f(u) = h(e^{-u})$ . Then  $f(u+v) = f(u) + f(v)$ . Continuity gives  $f(u) = au$ ; strict decrease of  $h$  gives  $a > 0$ .  $\square$

**Axiom 4.4** (Information-coherence naturality). *Let  $L = G^{-1}F$  on the positive information sector. The coherence generator is coordinate-free, commutes with experiment isomorphisms, depends only on  $L$ , and composes on direct sums.*

**Proposition 4.5** (Functional-calculus form). *Under information-coherence naturality and spectral locality, there exists a nonnegative Borel function  $\psi$  such that*

$$K = \psi(L). \quad (9)$$

*Proof.* Spectral locality assigns the same scalar action to equal spectral values, and conjugation naturality transports the assignment between eigenbases. The spectral theorem therefore represents the assignment by functional calculus. Direct-sum composition uses the same scalar function on each summand.  $\square$

**Axiom 4.6** (Antitonicity, homogeneity, and reciprocity). *For  $\lambda > 0$ , coherence loss obeys:*

$$\lambda_1 > \lambda_2 \implies \psi(\lambda_1) < \psi(\lambda_2), \quad (10)$$

$$\psi(a\lambda) = a^{-p}\psi(\lambda) \quad (a > 0), \quad (11)$$

$$\lambda\psi(\lambda) = \kappa \quad (12)$$

for a fixed normalization  $\kappa > 0$ .

**Theorem 4.7** (Reciprocal coherence constitutive law). *The preceding axioms uniquely imply*

$$K = \kappa L^{-1} \quad (13)$$

on the positive information sector.

*Proof.* Reciprocity directly gives  $\psi(\lambda) = \kappa/\lambda$ , which is strictly decreasing and homogeneous of degree  $-1$ . Functional calculus then gives  $K = \kappa L^{-1}$ .  $\square$

**Remark 4.8.** *The inverse is forced only after reciprocity is accepted. Naturality, monotonicity, and homogeneity alone leave the family  $\psi(\lambda) = c\lambda^{-p}$  with  $p > 0$ . Reciprocity is therefore a constitutive axiom, not a consequence of observation geometry.*

**Theorem 4.9** (Coherence-derived hard threshold). *Fix an observation horizon  $T > 0$  and minimum survival  $\varepsilon \in (0, 1)$ . With  $B = -a \log \varepsilon$ , the unique maximal coherence-surviving sector is*

$$W_{\text{coh}} = \text{im } \mathbf{1}_{[0, B]}(aTK). \quad (14)$$

Under  $K = \kappa L^{-1}$  this equals an upper information cut

$$W_{\text{coh}} = \text{im } \mathbf{1}_{[\tau_{\text{coh}}, \infty)}(L), \quad \tau_{\text{coh}} = \frac{a\kappa T}{B}. \quad (15)$$

Away from equality at the boundary, this is the RG hard projector convention after the endpoint convention is declared.

*Proof.* Mode survival is  $e^{-T\psi(\lambda)}$ . The condition  $-a \log c \leq B$  becomes  $aT\psi(\lambda) \leq B$ . Under reciprocity this is  $a\kappa T/\lambda \leq B$ , equivalently  $\lambda \geq a\kappa T/B$ . Spectral functional calculus gives the projector, and maximality retains every surviving mode.  $\square$

## 5 Stability and conditioning

**Definition 5.1** (Complete declared discriminant). *Let  $\Delta_\Gamma$  be the union of threshold crossings, information-rank boundaries, repeated-eigenvalue boundaries relevant to the declared invariant, nuisance-rank changes, conditioning failures, and transport singularities.*

**Definition 5.2** (Admissibility robustness radius). *For a declared perturbation norm and class  $\mathcal{U}$ , define*

$$r_\Gamma(x) = \inf\{\|\delta x\| : x + \delta x \in \Delta_\Gamma, \delta x \in \mathcal{U}\}. \quad (16)$$

**Theorem 5.3** (Canonical stability margin). *For fixed  $(\Delta_\Gamma, \mathcal{U}, \|\cdot\|)$ ,  $r_\Gamma$  is the unique distance-to-failure function. It is coordinate invariant under isometries of the declared norm, positive exactly off the closed discriminant, monotone under enlargement of the uncertainty set, and satisfies a minimum rule on independent direct sums when the product norm and discriminant decompose accordingly.*

*Proof.* All claims follow from standard properties of distance to a closed set and the definition of product distance. Numerical condition bounds may estimate this distance but do not equal it in general.  $\square$

**Corollary 5.4** (Operational stability). *Given a calibrated uncertainty radius  $\rho$ , the canonical robust-admissibility condition is*

$$r_{\Gamma}(x) > \rho. \quad (17)$$

*The value  $\rho$  remains experimental input.*

## 6 Nuisance closure

**Definition 6.1** (Generated nuisance relation). *Let  $\mathcal{H}_N$  be declared nuisance transformations. Define  $\sim_N$  as the smallest equivalence relation containing every pair  $(x, hx)$  for  $h \in \mathcal{H}_N$ .*

**Theorem 6.2** (Unique nuisance saturation). *The generated relation  $\sim_N$  exists and is unique. It is the intersection of all equivalence relations containing the declared generator pairs. If the generators form a group or groupoid action, its classes are the generated orbits.*

*Proof.* The universal relation is one containing relation, so the family is nonempty. Intersections of equivalence relations are equivalence relations. Their intersection is contained in every admissible relation and is therefore uniquely smallest.  $\square$

**Corollary 6.3.** *After nuisance generators are declared, no further choice remains in nuisance closure. Choosing the generators remains protocol input.*

## 7 Admissible coarse-graining

**Definition 7.1** (Generated information-order ideal). *Let  $\mathcal{M}$  be available parameter-independent garbling generators. The generated coarse-graining class  $C(\mathcal{M})$  is the smallest class containing identities and  $\mathcal{M}$  and closed under composition, declared products, and reachable Blackwell descendants.*

**Theorem 7.2** (Unique coarse-graining closure). *The generated class  $C(\mathcal{M})$  exists and is unique as the intersection of all classes satisfying the closure conditions. Equivalently in a finite garbling graph, it is the reachability closure of the declared generators.*

*Proof.* The full available morphism class is a closed containing class. Intersections preserve identity, composition, product, and downward-closure conditions. Hence the intersection is uniquely smallest. In finite graphs this is exactly transitive reachability.  $\square$

**Remark 7.3.** *Blackwell order constrains information loss under garbling; it does not assert that every mathematically possible garbling is physically available.*

## 8 Transport and the surviving modulus

**Axiom 8.1** (Compatible transport). *Transport preserves the declared comparison metric, intertwines  $P_{\tau}$  and  $W_{\text{obs|adm}}$ , is natural under experiment isomorphisms, and satisfies identity and cocycle laws.*

**Proposition 8.2** (Non-uniqueness of compatible connection). *Metric and projector compatibility do not determine a unique connection. If a resolved/null splitting has nontrivial skew endomorphisms within either block, then distinct block-skew connection generators preserve both the metric and projector.*

*Proof.* For  $G = I$  and  $P = \text{diag}(I_r, 0)$ , every block-diagonal skew operator  $A = \text{diag}(A_R, A_N)$  satisfies  $A^T + A = 0$  and  $AP = PA$ . Distinct choices generate distinct compatible transports.  $\square$

**Corollary 8.3** (Remaining open datum). *A unique connection requires an additional principle such as torsion removal in a setting where torsion is defined, least action, minimum curvature, or physical dynamical transport. Without it, the connection/holonomy class is irreducible protocol data.*

## 9 Conditional protocol characterization theorem

**Theorem 9.1** (Conditional characterization of the RG protocol). *Let  $\mathcal{E}$  be a regular finite classical experiment and let the generator package  $\Xi$  declare:*

1. *a strictly convex local decision/resource functional  $\mathcal{R}$ ;*
2. *a coherence normalization, horizon, and survival budget;*
3. *nuisance generators  $\mathcal{H}_N$ ;*
4. *a perturbation class, norm, uncertainty radius, and complete discriminant;*
5. *available garbling generators  $\mathcal{M}$ ;*
6. *optional connection-selection data  $\mathcal{V}$ .*

*Assume the Fisher module, coherence semigroup, spectral naturality, admissibility soundness and completeness, and the relevant closure axioms. If information-coherence reciprocity is declared, then:*

$$G = \nabla^2 \mathcal{R}, \tag{18}$$

$$K = \kappa(G^{-1}F)^{-1}, \tag{19}$$

$$\tau = \alpha \kappa T / B, \tag{20}$$

$$N = \text{the generated nuisance equivalence}, \tag{21}$$

$$S = \{x : r_\Gamma(x) > \rho\}, \tag{22}$$

$$C = C(\mathcal{M}). \tag{23}$$

*These components are unique relative to  $\Xi$ . The transport laws are characterized, but the connection is unique only if  $\mathcal{V}$  supplies a selection principle with a unique minimizer or universal solution.*

*Proof.* Combine the Hessian, coherence-clock, reciprocal constitutive, coherence-threshold, robustness-radius, nuisance-closure, and coarse-graining-closure theorems. Each component is unique relative to its declared generator. The compatible-connection proposition gives the precise exception.  $\square$

**Corollary 9.2** (Closure of the first protocol problem). *The structural form of  $G, \tau, N, S$ , and  $C$  is conditionally characterized. Their irreducible inputs are respectively the resource functional; coherence normalization/budget and reciprocity; nuisance generators; perturbation model and uncertainty radius; and available channels. The only component retaining intrinsic geometric moduli after these declarations is the compatible connection/holonomy class.*

## 10 Final battery

The deterministic attack used seed 20260806. Its modules produced:

| Module            | Result                                                                                                                                                                                                                                        |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constitutive law  | Among inverse powers, shifted inverse, exponential, and logarithmic candidates, only $\psi(\lambda) = \lambda^{-1}$ survived naturality, strict antitonicity, homogeneity, and reciprocity. Without reciprocity, all inverse powers survived. |
| Stability         | Distance to the complete discriminant satisfied boundary exactness, covariance, uncertainty monotonicity, and the direct-sum minimum rule. Condition number and threshold gap alone failed complete-boundary detection.                       |
| Nuisance          | Of 203 set partitions, the declared generators admitted one unique minimal saturated equivalence, with classes $\{0, 1, 2\}$ , $\{3, 4\}$ , and $\{5\}$ .                                                                                     |
| Coarse-graining   | The finite garbling graph admitted one unique maximal reachable downward-closed class from the declared generators.                                                                                                                           |
| Transport         | Four distinct metric- and projector-compatible skew connection generators survived, proving residual connection moduli.                                                                                                                       |
| Comparison metric | Among 256 diagonal candidate metrics, exactly one equaled the declared resource Hessian.                                                                                                                                                      |

**Remark 10.1.** *These computations do not prove the general theorems. They test finite implementations, expose axiom dependence, and provide countermodels to stronger protocol-free or connection-unique claims.*

## 11 Non-claims and falsification

**Non-claim 11.1** (No protocol from observation alone). *The generator package  $\Xi$  is not derived from  $\mathcal{E}$ .*

**Non-claim 11.2** (No reciprocity without a constitutive axiom). *The inverse law  $K = \kappa L^{-1}$  is not forced by naturality and monotonicity alone. Rejecting reciprocity restores a family of admissible constitutive laws.*

**Non-claim 11.3** (No absolute time or threshold). *The scale  $a$ , normalization  $\kappa$ , horizon  $T$ , and budget  $B$  require calibration or an external operational convention.*

**Non-claim 11.4** (No unique transport from compatibility). *Metric, projector, and cocycle compatibility do not eliminate connection and holonomy moduli.*

**Non-claim 11.5** (No automatic singular, infinite-dimensional, or quantum extension). *Closures, Hessians, inverses, discriminant distances, and semigroup generators require separate domain and regularity analysis outside the stated finite classical setting.*

The characterization requires revision if the declared Hessian is not positive; if the coherence family is not a semigroup; if reciprocity is empirically or structurally rejected; if the discriminant is not closed in the declared topology; if nuisance generators fail to act on the experiment; if available processing is not composition-closed; or if a claimed unique compatible connection admits distinct solutions.

## 12 Conclusion

The first RG protocol problem is not closed by deriving one universal protocol. It is closed structurally by separating generators from canonical closures. Once a local resource functional is declared, its Hessian fixes  $G$ . Once coherence semigroup, normalization, reciprocity, horizon, and budget are declared, spectral calculus fixes  $K$ ,  $\tau$ , and the hard survival sector. Nuisance generators fix their orbit closure. The perturbation model fixes distance to the discriminant. Available garblings fix their information-order closure. Transport compatibility fixes the admissible class but not its connection representative.

Accordingly, five protocol components are conditionally canonical, while transport retains a genuine modulus. This is the sharp endpoint supported by the theorem ladder and final battery.

## Reproducibility

Artifacts:

- `rg_protocol_closure_final_battery.py`;
- `rg_protocol_closure_final_battery.json`.

Seed: 20260806.

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